

# Jiawei Da

POSTDOCTORAL SCHOLAR

Department of Geosciences, The Pennsylvania State University

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## Research Interests

Isotope geochemistry; surface processes; biogeochemical cycles

## Education

### Nanjing University

Nanjing, China

DOCTOR OF PHILOSOPHY

09/2013-06/2020

- Dissertation: Quantitative reconstruction of paleoatmospheric CO<sub>2</sub> levels using pedogenic carbonates from the Chinese Loess Plateau
- Integrated Master's and PhD program
- Supervisor: Dr. Junfeng Ji

### Jilin University

Changchun, China

BACHELOR OF SCIENCE, GEOLOGY

09/2008-07/2012

## Research Experience

### The Pennsylvania State University

State College, PA

POSTDOCTORAL SCHOLAR

09/2025-Present

- Advisor: Dr. Max Lloyd
- Exploring the potential of methyl clumping as a biosignature

### The University of Texas at Austin

Austin, TX

POSTDOCTORAL FELLOW

08/2022-08/2025

- Advisor: Dr. Dan Breecker
- Modernizing paleosol-CO<sub>2</sub> reconstructions via clumped isotope and trace-amount organic carbon isotope analyses
- Reconstructing Rainfall seasonality over East Asia via clumped isotope and triple oxygen isotope of pedogenic carbonates
- Development of a Bayesian inversion framework of the paleosol proxy system model
- Understanding pedogenic carbonate formation in Vertisols through soil gas monitoring and modeling
- Identified dew uptake by land snails via Bayesian modeling

### Nanjing University

Nanjing, China

POSTDOCTORAL RESEARCH FELLOW

08/2020-07/2022

- Advisor: Dr. Xiancai Lu
- Unraveled pedogenic carbonate formation on the Chinese Loess Plateau using forward and numerical modeling
- Traced subsoil organic carbon turnover via radiocarbon fingerprinting

### Nanjing University

Nanjing, China

RESEARCH ASSISTANT

09/2015-07/2020

- Maintained the daily operation of FTIR and UV/VIS/NIR spectrometer

## Journal Articles

1. Okafor, B., **Da, J.**, Beverly, E. J., Driese, S., Nordt, L., & Breecker, D. O. (2025). A component of atmospheric vapor in the water of a floodplain Vertisol. *Journal of Hydrology*, 661, 133538. <https://doi.org/10.1016/j.jhydro1.2025.133538>
2. Mu, J., **Da, J.**, Yang, H., Ji, J., Liu, L., & Li, W. (2025). An abrupt change in source materials for Chinese loess deposits at the Pliocene-Pleistocene boundary: Insights from K isotopes and modeling. *Earth and Planetary Science Letters*, 668, 119543. <https://doi.org/10.1016/j.epsl.2025.119543>
3. **Da, J.** (2025). Isotopic evidence for dew uptake by land snails from Bayesian modeling of body water  $\delta D$  and  $\delta 18O$ . *Chemical Geology*, 692, 122953. <https://doi.org/10.1016/j.chemgeo.2025.122953>
4. **Da, J.**, Zhang, Y. G., Liu, X., Breecker, D. O., Li, G. K., Chen, T., & Ji, J. (2025). No apparent state-dependency of equilibrium climate sensitivity between the Pleistocene glacial and interglacial climate states. *Nature Communications*, 16(1), 6608. <https://doi.org/10.1038/s41467-025-61941-5>

5. Sakthivel, T., Ghosh, P., Nair, N., & **Da, J.** (2024). Wildfire-enhanced Plio-Pleistocene CO<sub>2</sub> drawdown through terrestrial organic carbon burial. *Quaternary Science Reviews*, 338, 108825. <https://doi.org/10.1016/j.quascirev.2024.108825>
6. **Da, J.**, Li, G. K., Breecker, D. O., & Ji, J. (2024). Particle-Size-Specific Radiocarbon Constraints Imply an Active Subsoil Organic Carbon Pool. *Journal of Geophysical Research: Biogeosciences*, 129(5), e2024JG008102. <https://doi.org/10.1029/2024JG008102>
7. THE CENOZOIC CO<sub>2</sub> PROXY INTEGRATION PROJECT (CENCO2PIP) CONSORTIUM. (2023). Toward a Cenozoic history of atmospheric CO<sub>2</sub>. *Science*, 382(6675), eadi5177. <https://doi.org/10.1126/science.adi5177>
8. **Da, J.**, Li, G. K., & Ji, J. (2023). Seasonal changes in the formation time of pedogenic carbonates on the Chinese Loess Plateau during Quaternary glacial cycles. *Quaternary Science Reviews*, 305, 108008. <https://doi.org/10.1016/j.quascirev.2023.108008>
9. **Da, J.**, Breecker, D. O., Li, T., Li, G., Lu, H., & Ji, J. (2023). A Humid East Asia During the Early Pliocene Indicated by Calcite Nodules From the Chinese Loess Plateau. *Paleoceanography and Paleoclimatology*, 38(7), e2023PA004615. <https://doi.org/10.1029/2023PA004615>
10. Bao, R., Sheng, X., Meng, X., Li, T., Li, C., Shen, H., **Da, J.**, Ji, J., & Chen, J. (2022). 100 k.y. Pacing of the East Asian summer monsoon over the past five glacial cycles inferred from land snails. *Geology*. <https://doi.org/10.1130/G50243.1>
11. Meng, X., Li, G. K., Liu, L., Long, X., Zhao, W., **Da, J.**, & Ji, J. (2022). Decoupled paleosol-based proxies in Chinese loess deposits: Role of leaching and illuviation processes. *Quaternary Science Reviews*, 298, 107847. <https://doi.org/10.1016/j.quascirev.2022.107847>
12. **Da, J.**, Li, G. K., & Ji, J. (2021). Overestimate of C<sub>4</sub> Plant Abundance Caused by Soil Degradation-Induced Carbon Isotope Fractionation. *Geophysical Research Letters*, 48(24), e2021GL093407. <https://doi.org/10.1029/2021GL093407>
13. **Da, J.**, Zhang, Y. G., Li, G., & Ji, J. (2020). Aridity-driven decoupling of  $\delta^{13}\text{C}$  between pedogenic carbonate and soil organic matter. *Geology*. <https://doi.org/10.1130/G47241.1>
14. **Da, J.**, Zhang, Y. G., Li, G., Meng, X., & Ji, J. (2019). Low CO<sub>2</sub> levels of the entire Pleistocene epoch. *Nature Communications*, 10(1), 4342. <https://doi.org/10.1038/s41467-019-12357-5>
15. **Da, J.**, Zhang, Y. G., Wang, H., Balsam, W., & Ji, J. (2015). An Early Pleistocene atmospheric CO<sub>2</sub> record based on pedogenic carbonate from the Chinese loess deposits. *Earth and Planetary Science Letters*, 426, 69–75. <https://doi.org/10.1016/j.epsl.2015.05.053>

## Under Review

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1. Chen, Z., **Da, J.**, Li, C., Xue, L., Liu, G., Sheng, X., Huang, J., & Ji, J. (2025). *Crystallization fractionation in anthropogenic carbonate: Geochemical constraints on dead carbon*. SCIENCE CHINA Earth Sciences.
2. Czwakiel, N., Gallagher, T., Serach, L., Ludvigson, G., Gao, P., Nie, J., Suc, J.-P., **Da, J.**, & Breecker, D. (2025). *The onset of summer drying on the iberian peninsula during pliocene global cooling*. *Paleoceanography and Paleoclimatology*.
3. **Da, J.**, Bowen, G., Harper, D., & Huntington, K. (2025). *Improving paleosol-based atmospheric CO<sub>2</sub> reconstruction via joint proxy inversion*. *Paleoceanography; Paleoclimatology*.
4. Li, C., Bao, R., **Da, J.**, & Sheng, X. (2025). *Formation processes control on  $\delta^{13}\text{C}$  and  $\delta^{18}\text{O}$  between biogenic vs. Non-biogenic carbonates in loess-paleosol sections*. *Journal of Asian Earth Sciences*.

## In Preparation

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1. **Da, J.**, Beverly, E., Driese, S., & Breecker, D. (2025). *Elevated soil respiration with depth in vertisols regulated by soil porosity*. In prep for *Geochimica et Cosmochimica Acta*.
2. **Da, J.**, Serach, L., Sun, C., Gallagher, T., Lu, H., Zhang, H., Wang, H., Ji, S., Huntington, K., Sharp, Z., Feng, R., Ji, J., & Breecker, D. (2025). *Jet-induced rainfall seasonality and C<sub>4</sub> expansion over east asia*. In prep for *Nature*.

## Invited Talks

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### Isotopic evidence of dew uptake by land snails

THE UNIVERSITY OF TEXAS AT AUSTIN

2025

### Isotopic evidence of dew uptake by land snails

INSTITUTE OF GEOLOGY AND GEOPHYSICS, CHINESE ACADEMY OF SCIENCES

2025

### Late Miocene-Pliocene rainfall seasonality over East Asia through the lens of the westerly jet

UNIVERSITY OF NEVADA, LAS VEGAS

2025

### Miocene-Pliocene rainfall seasonality over East Asia recorded by soil carbonates

SOUTHERN METHODIST UNIVERSITY

2025

### How to make the paleosol-CO<sub>2</sub> proxy more useful?

SOUTHERN METHODIST UNIVERSITY

2025

### East Asian hydroclimate during the Pliocene: new isotopic evidence from soil carbonate

UNIVERSITY OF WASHINGTON

2024

### Continual glacial CO<sub>2</sub> drawdown recorded by paleosols from the Chinese Loess Plateau

THE UNIVERSITY OF NEW MEXICO

2023

### Reconstructing past atmospheric CO<sub>2</sub> levels with pedogenic carbonates from the Chinese loess deposits

THE UNIVERSITY OF TEXAS AT AUSTIN

2022

## Conference Presentations

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1. **Da, J.**, Serach, L., Gallagher, T., Sun, C., Lu, H., Zhang, H., Wang, H., Ji, S., Huntington, K., Sharp, Z., Feng, R., Ji, J., & Breecker, D. (2024, December). *Pliocene hydroclimate changes over eastern china through the lens of the westerlies*. [Poster]. AGU fall meeting. Washington DC, DC, USA.
2. **Da, J.**, Serach, L., Gallagher, T., Sun, C., Lu, H., Zhang, H., Wang, H., Ji, S., Huntington, K., Sharp, Z., Feng, R., Ji, J., & Breecker, D. (2024, August). *Summer drought over east asia during the warm pliocene: Evidence from clumped isotope and triple oxygen isotope compositions of soil carbonate*. [Talk]. 2024 goldschmidt conference. Chicago, IL, USA.
3. **Da, J.**, Serach, L., Gallagher, T., Sun, C., Lu, H., Zhang, H., Wang, H., Ji, S., Huntington, K., Sharp, Z., Feng, R., Ji, J., & Breecker, D. (2024, April). *Enhanced summer drought over east asia across the miocene-pliocene boundary*. [Poster]. Miocene climate workshop. Tucson, AZ, USA.
4. **Da, J.**, Zhang, Y. G., Liu, X., Li, G., Breecker, D., & Ji, J. (2023, December). *Pleistocene global cooling driven by declining glacial CO<sub>2</sub> levels*. [Invited talk]. AGU fall meeting. San francisco, CA, USA.
5. Bowen, G., Harper, D., **Da, J.**, Hönisch, B., & Montañez, I. (2023, October). *Toward an omni-proxy reconstruction of cenozoic CO<sub>2</sub>*. [Talk]. The geological science of america meeting. Pittsburgh, PA, USA.
6. **Da, J.**, Breecker, D., Lu, H., & Ji, J. (2023, October). *A humid east asia during the early pliocene indicated by calcite nodules from the chinese loess plateau*. [Invited talk]. The geological science of america meeting. Pittsburgh, PA, USA.
7. **Da, J.**, Li, G., & Ji, J. (2023, July). *Seasonal changes in the formation time of pedogenic carbonates on the chinese loess plateau during quaternary glacial cycles*. [Talk]. 2024 goldschmidt conference. Leon, france.
8. **Da, J.**, Zhang, Y. G., Li, G., & Ji, J. (2022, November). *Reconstructing pleistocene atmospheric CO<sub>2</sub> levels using pedogenic carbonates from the chinese loess plateau*. [Talk]. INQUA LoessFest. Virtual conference.
9. **Da, J.**, Li, G., & Ji, J. (2021, November). *Carbon isotope fractionation during the burial and decomposition of soil organic matter – evidence from the paleosols on the chinese loess plateau*. [Talk]. 8th biology and organic geochemistry conference. Xiamen, china.
10. **Da, J.**, Zhang, Y. G., Li, G., & Ji, J. (2021, August). *Quantitative constraint of the effect of atmospheric CO<sub>2</sub> on the isotopic compositions of pedogenic carbonates on the chinese loess plateau*. [Talk]. The 6th conference on earth system science. Shanghai, china.

11. **Da, J.**, Zhang, Y. G., Li, G., & Ji, J. (2020, August). *Refining the paleosol-CO<sub>2</sub> proxy and the reconstruction of early-pleistocene CO<sub>2</sub> levels*. [Talk]. 2020 goldschmidt conference. Virtual conference.
12. **Da, J.**, & Ji, J. (2016, July). *Reconstructing past atmospheric CO<sub>2</sub> levels with pedogenic carbonates from the chinese loess deposits*. [Poster]. 2016 goldschmidt conference. Yokohama, japan.

## Awards

|                                                                                                                              |         |
|------------------------------------------------------------------------------------------------------------------------------|---------|
| <b>NSF-China Earth Sciences Postdoctoral Fellowship</b>                                                                      | 2021    |
| NATIONAL SCIENCE FOUNDATION OF CHINA                                                                                         | \$15000 |
| <b>Best Doctoral Dissertation Award</b>                                                                                      | 2021    |
| JIANGSU PROVINCIAL DEPARTMENT OF EDUCATION                                                                                   |         |
| <b>Best Doctoral Dissertation Award</b>                                                                                      | 2021    |
| NANJING UNIVERSITY                                                                                                           |         |
| <b>Yuxiu Young Scholar Program Postdoctoral Fellowship</b>                                                                   | 2020    |
| NANJING UNIVERSITY                                                                                                           | \$86000 |
| <b>Li Siguang Outstanding PhD Candidate Award</b>                                                                            | 2020    |
| LI SIGUANG GEOLOGICAL SCIENCE AWARD FOUNDATION                                                                               | \$4300  |
| • National award to five selective Ph.D. candidates majored in Geology per year in recognition of high academic achievements |         |
| <b>Outstanding PhD student</b>                                                                                               | 2020    |
| NANJING UNIVERSITY                                                                                                           |         |
| <b>Program A for outstanding Ph.D. students, Nanjing University</b>                                                          | 2018    |
| NANJING UNIVERSITY                                                                                                           | \$17000 |
| <b>First Prize of National Scholarship</b>                                                                                   | 2015    |
| MINISTRY OF EDUCATION                                                                                                        | \$4200  |

## Grant-writing Experience

|                                                                                                                                                         |                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| <b>Reconstruction of Holocene paleoenvironmental gradients across the southern Great Plains using snailshells from archaeological collections</b>       | 2024              |
| NATIONAL SCIENCE FOUNDATION                                                                                                                             | Not Funded        |
| • Role: Formal analysis; reviewing and editing                                                                                                          |                   |
| <b>Quantifying the decomposition-related carbon isotopic fractionation of soil organic matter in the eolian deposits from the Chinese Loess Plateau</b> | 2021              |
| NATIONAL NATURAL SCIENCE FOUNDATION OF CHINA                                                                                                            | Funded (\$41000)  |
| • Role: Conceptualization; original draft preparation; formal analysis; reviewing and editing                                                           |                   |
| <b>Understanding the seasonality and formation of pedogenic carbonate on the Chinese Loess Plateau</b>                                                  | 2021              |
| CHINA POSTDOCTORAL SCIENCE FOUNDATION                                                                                                                   | Funded (\$7000)   |
| • Role: Conceptualization; original draft preparation; formal analysis; reviewing and editing                                                           |                   |
| <b>Reconstructing atmospheric CO<sub>2</sub> levels over the past eight million years using the eolian deposits from the Chinese Loess Plateau</b>      | 2020              |
| NATIONAL NATURAL SCIENCE FOUNDATION OF CHINA                                                                                                            | Funded (\$400000) |
| • Role: Conceptualization; original draft preparation; formal analysis; reviewing and editing                                                           |                   |
| <b>Evaluating atmospheric CO<sub>2</sub> signal in the carbon isotope composition of calcite nodules from the Chinese Loess Plateau</b>                 | 2018              |
| NATIONAL NATURAL SCIENCE FOUNDATION OF CHINA                                                                                                            | Funded (\$92000)  |
| • Role: original draft preparation; formal analysis; reviewing and editing                                                                              |                   |

## Small Grants

|                                   |        |
|-----------------------------------|--------|
| <b>JSG Go Further Find</b>        | 2024   |
| THE UNIVERSITY OF TEXAS AT AUSTIN | \$1000 |

|                                                        |        |
|--------------------------------------------------------|--------|
| <b>Miocene Climate Workshop Travel Grant</b>           | 2024   |
| NATIONAL SCIENCE FOUNDATION                            | \$1000 |
| <b>UT Staff Council Professional Development Grant</b> | 2023   |
| THE UNIVERSITY OF TEXAS AT AUSTIN                      | \$1500 |
| <b>Goldschmidt Travel Grant</b>                        | 2016   |
| GEOCHEMICAL SOCIETY                                    | \$1000 |

## Teaching Experience

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### GEO 401: Physical Geology

2024 Fall

LECTURER

- Participated in course design; Taught weekly lectures

### GEO 391: Environmental Isotope Geochemistry

2025 Spring

CO-INSTRUCTOR

- Participated in course design; Taught weekly lectures

## Professional Service

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### CONFERENCE SESSION CONVENOR

- AGU Fall Meeting, December 2023, San Francisco, CA, US

### UNDERGRADUATE/HIGH SCHOOL STUDENT MENTOR

- 2016 - current, mentored undergraduate students conducting research projects.
- Mentored students: Jinjin Yang (2016-2017), Xia Wang (2018-2019), Ruiqing Ji (2021-2022), Hudson Thomas (2024)

### GRADUATE MENTOR

- 2018 - current, mentored graduate students conducting research projects.
- Mentored students: Hanzhao Zhai (2018-), Jun Mu (2020-2022), Chenglong Li (2018-2025), Zhanpeng Chen (2022-), Nicole Czwakiel (2022-2025), Morgan Mllum (2024-)

### JUDGE

- 2025, Longhorn Research Poster Session
- 2024, OSPA Judge, AGU Fall Meeting
- 2023, the 13th Annual Jackson School of Geoscience Student Research Symposium
- 2023, OSPA Judge, AGU Fall Meeting

### AD HOC REVIEWER

- Science Advances
- Proceedings of the National Academy of Sciences (2)
- Earth Science Reviews
- Geophysical Research Letters
- Paleoceanography and Paleoclimatology (2)
- Global and Planetary Change (2)
- Earth's Future
- Chemical Geology (2)
- Quaternary Science Reviews
- Palaeogeography, Palaeoclimatology, Palaeoecology (2)
- Geochemistry, Geophysics, Geosystems
- Geoderma
- Environmental Science & Technology
- Applied Geochemistry
- Atmosphere (3)

- Arabian Journal of Geosciences
- Vertebrate Paleobiology and Paleoanthropology Series
- Water
- Forests
- Soil systems

## Skill Sets

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### LAB TECHNIQUES

- Isotope Ratio Mass Spectrometer (MAT253, Nucarb)
- Laser Spectroscopy (Picarro CRDS, Aerodyne TILDAS)
- Elemental Analyzer (EA)
- Fourier Transform Infrared Spectroscopy (FTIR)
- Scanning Electronic Microscopy (SEM)

### PROGRAMMING AND SOFTWARE

- Rstudio (data analyses, visualization, modeling, mapping)
- Excel (data analyses, visualization)
- CorelDRAW (visualization)

### LANGUAGES

- Chinese (Native Speaker)
- English (Professional proficiency)

## References

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### **Junfeng Ji, Ph.D.**

PROFESSOR, NANJING UNIVERSITY

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### **Daniel Breecker, Ph.D.**

PROFESSOR, THE UNIVERSITY OF TEXAS AT AUSTIN

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### **Gabe Bowen, Ph.D.**

DEPARTMENT CHAIR, PROFESSOR, UNIVERSITY OF UTAH

*gabe.bowen@utah.edu*

### **Katharine Huntington, Ph.D.**

PROFESSOR, UNIVERSITY OF WASHINGTON

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### **Isabel Montañez, Ph.D.**

DIRECTOR, INSTITUTE OF THE ENVIRONMENT, DISTINGUISHED PROFESSOR, UC DAVIS

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### **Max Lloyd, Ph.D.**

ASSISTANT PROFESSOR, PENNSYLVANIA STATE UNIVERSITY

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